■ AI RESEARCH OPERATING SYSTEM · VERSION 3.0 ■

KIMI K2.5
SWARM DEEP RESEARCH
PROMPT SYSTEM

A Full-Scale Multi-Agent Research Architecture for Deep Analytical Intelligence

■ Command Core Architecture	■ 6-Agent Swarm System	■ Recursive Refinement
■■ Bias Detection Protocol	■ Precision Output Formatter	■ Cross-Domain Synthesis

Deploy this system inside Kimi K2.5, Claude, GPT-4o, or any frontier LLM for elite-grade intelligence gathering across any domain.

◆ 2026 EDITION ◆ FULLY UPGRADED ◆ REUSABLE FOR ANY TOPIC ◆

NAVIGATION

TABLE OF CONTENTS

01 SYSTEM OVERVIEW & PHILOSOPHY 01
- What Makes This Different
- Deployment Context
- Core Principles

02 MASTER SWARM PROMPT (COMPLETE) 02
- Full Deployable Prompt
- Variable Substitution Guide
- Activation Triggers

03 AGENT ARCHITECTURE 03
- Command Core Agent
- Intelligence Expansion Agents
- Data Mining Agents
- Critical Challenge Agent
- Strategic Synthesis Agent
- Precision Formatter Agent

04 DEPTH PROTOCOL 04
- 3-Layer Deep Dive Framework
- Mechanism Extraction
- What-If Analysis

05 OUTPUT STRUCTURE SYSTEM 05
- All 10 Output Formats
- Formatting Specifications

■■ Compression Layers

06 VERIFICATION & BIAS DETECTION 06

■■ Source Validation
■■ Logical Fallacy Detection
■■ Cross-Agent Verification

07 RECURSIVE REFINEMENT FRAMEWORK 07

■■ Iterative Deepening
■■ Self-Correction Loop
■■ Quality Gates

08 IMPLEMENTATION GUIDE 08

■■ Step-by-Step Deployment
■■ Topic Customization
■■ Advanced Configuration

■ SECTION 01

SYSTEM OVERVIEW & PHILOSOPHY

This is not a prompt. This is an **AI Research Operating System**. It simulates a distributed multi-agent swarm where specialized cognitive modules collaborate, challenge each other, and synthesize intelligence far beyond what any single-pass query can produce.

WHAT MAKES THIS DIFFERENT

STANDARD PROMPT	THIS SWARM SYSTEM
Single agent, single pass	6 specialized agents in coordinated architecture
Surface-level answers	3-layer deep mechanism extraction
No self-correction	Recursive refinement with quality gates
Accepts information at face value	Active contradiction & bias detection
One output format	10 configurable output formats
Topic drift frequent	Command Core enforces scope at every step
No source validation	Multi-layer source credibility protocol

CORE DESIGN PRINCIPLES

- **Depth Over Breadth**: Every agent is instructed to go 3 layers deeper than surface explanations. Mechanism-level understanding is the minimum acceptable output.
- **Adversarial Collaboration**: Agents are designed to challenge each other's outputs. The Critical Challenge Agent's sole purpose is to break assumptions before they reach the synthesis layer.
- **Structured Parsability**: All outputs follow rigid XML-like tagging so downstream LLMs, humans, or automation pipelines can extract specific intelligence nodes.
- **Bias Immunity Protocol**: Geographic, recency, confirmation, and authority biases are explicitly flagged and corrected before synthesis.
- **Compression-Ready**: Every deep output is automatically compressed into 3 tiers — full report, 1-page summary, and ultra-compressed memory-saving version.
- **Universal Applicability**: The system is topic-agnostic. Replace [TOPIC] with any domain: finance, science, geopolitics, technology, medicine, law, or creative fields.

DEPLOYMENT CONTEXT

This system is optimized for Kimi K2.5 Agent Swarm mode (100 parallel sub-agents, 1,500 tool calls, 256K context window), but is fully deployable in any frontier LLM including Claude Opus, GPT-4o, Gemini Ultra, or Llama-3 variants. The architecture adapts automatically — on single-agent models, the swarm becomes a sequential multi-pass protocol achieving the same depth through iteration.

■ SECTION 02

MASTER SWARM PROMPT (COMPLETE)

This is the complete, deployable prompt. Copy it in full. Replace all [BRACKETED VARIABLES]. Send it as your opening message. The system activates automatically.

FULL DEPLOYABLE MASTER PROMPT

■ LAYER 1 — MISSION DECLARATION

```
You are now operating as a Kimi K2.5 Swarm Deep Research System.

Your task: Conduct a comprehensive multi-agent deep research session on [EXACT TOPIC].

MISSION OBJECTIVE: [SPECIFIC OUTCOME — what decision, document, or insight this
produces]

RESEARCH PURPOSE: [CONTEXT — competitive analysis / investment memo / policy brief /
R&D; report]

STAKES LEVEL: [HIGH / MEDIUM / LOW] — calibrate depth and verification rigor
accordingly.

PRIMARY AUDIENCE: [WHO READS THIS — executives / researchers / engineers / investors]

SUCCESS DEFINITION: Research is complete when every sub-question in Layer 4 has been

answered at mechanism level with supporting evidence and confidence scores.
```

■ LAYER 2 — SCOPE & BOUNDARIES

```
GEOGRAPHIC FOCUS: [Countries / Regions / Global]

TIME PERIOD: [Date range — e.g., 2022–2025 only]

SOURCE TYPES INCLUDE: [Academic papers / news / corporate filings / patents /
interviews]

SOURCE TYPES EXCLUDE: [Social media speculation / opinion blogs / unverified reports]

LANGUAGE PRIORITY: English first, then [secondary languages if relevant]

DEPTH CEILING: Go maximum 3 sub-layers deep on each domain. Do not exceed unless

a finding in Layer 2 reveals a critical unknown that changes the mission scope.

SCOPE LOCK: If any agent detects topic drift, halt and return to Mission Objective.
```

■ LAYER 3 — AGENT BEHAVIORAL CONSTRAINTS

```
NEGATIVE SPACE RULES (what agents must NOT do):
```

✗ Do not resolve contradictions — flag them and surface them in the output.

✗ Do not fill gaps with inference — label all inferred content [INFERRED].

✗ Do not spawn additional research loops without flagging the trigger reason.

✗ Do not assign confidence >85% without 3+ independent corroborating sources.

✗ Do not suppress minority expert positions — include dissenting views explicitly.

✗ Do not prioritize recency over accuracy — flag when older data is more reliable.

✗ Do not merge agent outputs that contain unresolved contradictions.

■ LAYER 4 — WORKSTREAM DECOMPOSITION

Spawn the following parallel research workstreams simultaneously:

WORKSTREAM A: [Core Domain Analysis] — Mechanism, history, current state

WORKSTREAM B: [Quantitative Landscape] — Data, metrics, trends, benchmarks

WORKSTREAM C: [Actor/Stakeholder Map] — Key players, motivations, power dynamics

WORKSTREAM D: [Risk & Failure Analysis] — Scenarios, vulnerabilities, edge cases

WORKSTREAM E: [Innovation/Opportunity Frontier] — Emerging developments, whitespace

WORKSTREAM F: [Regulatory/Structural Context] — Governance, constraints, norms

Each workstream must return: Findings / Evidence / Confidence Score / Open Questions

■ LAYER 5 — AGENT INTERACTION TOPOLOGY

CROSS-VERIFICATION PROTOCOL:

· Workstream B must validate all quantitative claims in Workstream A.

· Workstream C must assess how actor incentives affect Workstream D scenarios.

· Workstream F must flag regulatory constraints on Workstream E opportunities.

· Critical Challenge Agent reviews ALL workstream outputs before synthesis.

CONFLICT ESCALATION RULE:

If two workstreams produce contradictory findings, both versions are preserved.

Tag conflicting pairs as [CONFLICT-UNRESOLVED] and include in executive summary.

■ LAYER 6 — SOURCE VALIDATION PROTOCOL

SOURCE TIER SYSTEM:

TIER 1 (Weight: HIGH): Peer-reviewed papers, government data, audited financial reports

TIER 2 (Weight: MEDIUM): Established news organizations, industry analysts, patents

TIER 3 (Weight: LOW): Expert commentary, conference proceedings, white papers

```
TIER 4 (EXCLUDED): Anonymous sources, social media, unattributed claims

BIAS CORRECTION MANDATES:
· Flag geographic bias: Check if Western/English sources over-represent findings.
· Flag recency bias: Validate that recent ≠ more important in this context.
· Flag authority bias: Cross-check consensus positions against dissenting research.
```

■ LAYER 7 — QUALITY & OUTPUT STANDARDS

```
MINIMUM QUALITY GATES (each workstream must pass before synthesis):
✓ Every factual claim has a source citation or [INFERRED] tag
✓ Confidence scores assigned: HIGH (>75%) / MEDIUM (50–75%) / LOW (<50%)
✓ Contradictions documented — not resolved without explicit evidence
✓ At least 1 failure scenario per major finding
✓ At least 1 alternative explanation per key conclusion

OUTPUT FORMAT REQUESTED: [CHOOSE FROM SECTION 05 OF THIS GUIDE]
COMPRESSION LEVEL: [FULL / 1-PAGE / ULTRA-COMPRESSED]
```

VARIABLE SUBSTITUTION GUIDE

VARIABLE	WHAT TO ENTER	EXAMPLE
[EXACT TOPIC]	Precise research subject	"CRISPR gene editing in oncology 2024"
[SPECIFIC OUTCOME]	Deliverable type	"Investor memo on market opportunity"
[CONTEXT]	Use case	"Board presentation for Series B raise"
[STAKES LEVEL]	HIGH / MEDIUM / LOW	HIGH = max verification rigor
[PRIMARY AUDIENCE]	Reader profile	"Non-technical C-suite executives"
[Date range]	Research window	"January 2022 – December 2024 only"
[OUTPUT FORMAT]	One of 10 formats	"Deep Report + Executive Brief"

■ SECTION 03

AGENT ARCHITECTURE

The swarm consists of 6 specialized cognitive agents. Each has a fixed role, strict behavioral rules, and defined interaction protocols with other agents. No agent operates in isolation.

■ AGENT 1: COMMAND CORE

Orchestrator · Scope Enforcer · Direction Controller

- Reads the Mission Declaration and initializes the agent schema before any research begins
- Defines non-negotiable scope boundaries — prevents topic drift across all workstreams
- Monitors tool call budget and reallocates resources based on emerging findings
- Halts any workstream that deviates from scope and reissues corrected instructions
- Makes final arbitration decisions when agents produce conflicting recommendations
- Ensures depth > breadth at every decision point — will always trade coverage for precision
- Generates the session's master knowledge tree before passing to Synthesis Agent

■ AGENT 2: INTELLIGENCE EXPANSION

Domain Mapper · Hierarchy Builder · Cross-Domain Scout

- Breaks the topic into sub-domains and maps conceptual hierarchies (3 levels deep minimum)
- Identifies non-obvious relationships between the topic and adjacent fields
- Detects cross-disciplinary leverage points — where insights from Field A solve problems in Field B
- Builds the conceptual map that all other agents use as their navigation reference
- Flags known unknowns: gaps in current knowledge that are themselves meaningful signals
- Identifies which sub-domains are over-researched vs. under-researched globally

■■ AGENT 3: DATA MINING

Signal Extractor · Pattern Detector · Contradiction Surfacer

- Extracts both quantitative signals (statistics, benchmarks, growth rates) and qualitative signals (expert consensus, cultural factors, narrative shifts)
- Identifies recurring patterns across multiple independent sources — high-confidence indicators
- Actively surfaces contradictions: findings that conflict with mainstream positions
- Ranks insights by: (1) Novelty, (2) Evidence strength, (3) Strategic relevance
- Applies source tier weighting — Tier 1 sources receive 3x weight vs. Tier 3
- Tags all data points with recency flags and known data quality issues

- Returns structured data in: [FINDING] | [SOURCE TIER] | [CONFIDENCE] | [CONTRADICTS:] format

■■ AGENT 4: CRITICAL CHALLENGE

Assumption Attacker · Fallacy Detector · Stress-Tester

- Attacks every major claim from Agents 2 and 3 before they reach synthesis
- Detects and labels: straw man, false dichotomy, hasty generalization, appeal to authority
- Evaluates counterarguments from credible opposing positions — devil's advocate at maximum strength
- Stress-tests each conclusion against: worst-case scenario, delayed timeline, resource constraints
- Checks for omission bias: what has been left out that would change the conclusion?
- Evaluates whether findings hold across different cultural or geographic contexts
- Returns a RED / AMBER / GREEN rating for each major claim: RED = insufficient evidence, AMBER = requires caveat, GREEN = well-supported

■ AGENT 5: STRATEGIC SYNTHESIS

Output Merger · Redundancy Remover · Knowledge Tree Builder

- Receives outputs from all other agents and performs structured merging
- Removes redundancy while preserving nuance — two sources saying the same thing ≠ double confidence
- Builds the final knowledge tree: hierarchical structure from macro insights to micro evidence
- Ensures all [CONFLICT-UNRESOLVED] tags are preserved and surfaced prominently
- Generates executive-grade conclusions: 3 strategic implications, 3 immediate actions, 3 open questions
- Cross-references Critical Challenge Agent's ratings — no RED-rated claim appears as a conclusion
- Produces the master synthesis document that Precision Formatter receives as input

■ AGENT 6: PRECISION FORMATTER

Output Architect · Multi-Format Generator · Compression Engine

- Converts Synthesis Agent output into the requested format with zero information loss
- Generates all required output tiers: Full Report → 1-Page Summary → Ultra-Compressed
- Applies strict visual hierarchy: sections, sub-sections, callouts, tables, evidence blocks
- Ensures LLM parseability: every section is tagged, labeled, and consistently structured
- Inserts confidence scores, source citations, and [INFERRED] labels in standardized positions
- Generates the Conceptual Map as a structured text hierarchy if visual rendering is unavailable

- Final quality check: verifies all 10 quality gates are passed before returning output

SECTION 04

DEPTH PROTOCOL

Surface answers are noise. The Depth Protocol forces every research thread to descend three layers below the obvious — from what, to how, to why, to what-if.

THE 3-LAYER DEPTH FRAMEWORK

LAYER 1: WHAT (Surface)

- Definition and description of the phenomenon
- Current state: who, what, where, when
- Standard narrative — what most sources agree on
- Key statistics and headline metrics

LAYER 2: HOW (Mechanism)

- Causal chains: what produces the observed phenomenon
- System dynamics: feedback loops, dependencies, constraints
- Process-level explanation: step-by-step mechanism
- Enabling conditions: what must be true for this to occur

LAYER 3: WHY + WHAT-IF (Deep Structure)

- Underlying incentive structures and root causes
- Second-order effects: what happens downstream if this changes
- Counterfactual analysis: what would be different if X were false
- Failure scenarios: under what conditions does this break
- Optimization pathways: what would make this better / worse
- Paradigm dependency: which assumptions, if changed, collapse this conclusion

FAILURE SCENARIO PROTOCOL

For every major finding, the system must generate at minimum one failure scenario across each of the following failure modes:

FAILURE MODE	TRIGGER CONDITION	DETECTION METHOD
Resource Failure	Critical dependency removed or priced out	Sensitivity analysis on key inputs
Timeline Failure	Projected outcome delayed by 3–5x	Historical base rate comparison

Assumption Failure	Core premise shown to be false	Counterfactual stress test
Black Swan	Low-probability, high-impact disruption	Tail risk scenario modeling
Coordination Failure	Multi-actor dependency breaks down	Game theory / incentive mapping
Regulatory Failure	Policy environment shifts adversely	Regulatory horizon scanning

■ SECTION 05

OUTPUT STRUCTURE SYSTEM

ALL 10 OUTPUT FORMATS

01	EXECUTIVE SUMMARY	High-impact 3-paragraph brief. Strategic implications only. No methodology. Decision-ready.
02	CONCEPTUAL MAP	Hierarchical text structure showing topic → sub-domains → mechanisms → evidence nodes.
03	DEEP DOMAIN ANALYSIS	Full mechanism-level exploration of each workstream. 3-layer depth. With evidence citations and confidence scores.
04	CROSS-LINKING INSIGHTS	Non-obvious connections between workstreams. Where finding from Domain A changes conclusions in Domain B.
05	CONTRADICTIONS & EDGE CASES	All unresolved conflicts. Minority expert positions. Edge cases that break the mainstream narrative.
06	STRATEGIC IMPLICATIONS	3–5 specific, actionable strategic moves derived from the research. With risk/reward assessment.
07	FUTURE OUTLOOK	3 scenarios (optimistic / base / pessimistic) with probability estimates and key trigger events.
08	PRACTICAL IMPLEMENTATION	Step-by-step action blueprint. Who does what, when, in what sequence, with what resources.
09	1-PAGE COMPRESSION	Full research compressed to exactly 1 page. Preserves all critical insights. Sacrifices nothing essential.
10	ULTRA-COMPRESSED	5-bullet maximum. Each bullet = 1 sentence. For memory-saving or rapid briefing scenarios.

STANDARD SECTION TAGGING

All outputs must use standardized XML-style tags for machine parseability:

■ REQUIRED SECTION TAGS

```
[EXEC-SUMMARY] ... [/EXEC-SUMMARY]
```

```
[CONCEPT-MAP] ... [/CONCEPT-MAP]
[DOMAIN-ANALYSIS: workstream_name] ... [/DOMAIN-ANALYSIS]
[CROSS-LINK: domain_a ↔ domain_b] ... [/CROSS-LINK]
[CONFLICT-UNRESOLVED: claim_description] ... [/CONFLICT]
[STRATEGIC-IMPLICATION: priority_rank] ... [/STRATEGIC]
[SCENARIO: optimistic/base/pessimistic] ... [/SCENARIO]
[ACTION-ITEM: owner, timeline, resource] ... [/ACTION]
[CONFIDENCE: HIGH/MEDIUM/LOW] ... [/CONFIDENCE]
[INFERRED: basis_for_inference] ... [/INFERRED]
```

■ SECTION 06

VERIFICATION & BIAS DETECTION

Every insight is only as reliable as its weakest source and most unchallenged assumption. This protocol systematically eliminates the most common failure modes in AI research.

SOURCE VALIDATION MATRIX

SOURCE TYPE	TIER	WEIGHT	REQUIRED CORROBORATION
Peer-reviewed journals	TIER 1	3×	None — cite and use
Government/official statistics	TIER 1	3×	Cross-check for data vintage
Audited financial reports	TIER 1	3×	Verify reporting period
Established news organizations	TIER 2	1.5×	1 additional corroborating source
Industry analyst reports	TIER 2	1.5×	Check analyst incentives
Patent filings	TIER 2	1.5×	Verify approval status
Expert commentary/interviews	TIER 3	1×	2 additional sources required
Conference papers	TIER 3	1×	Check peer review status
Social media / forums	TIER 4	0×	EXCLUDED — never cite directly
Anonymous sources	TIER 4	0×	EXCLUDED — flag if it's the only source

BIAS DETECTION CHECKLIST

Geographic Bias	Are 80%+ of sources from one region? If yes, flag and search for non-Western perspectives.
Recency Bias	Are older foundational studies being ignored in favor of recent noise? Validate both.
Confirmation Bias	Is the research only finding evidence that supports the hypothesis? Actively search for disconfirming evidence.
Authority Bias	Is consensus being accepted because of who said it, not what evidence supports it?
Availability Bias	Are easily findable sources over-weighted vs. harder-to-access but more reliable data?

Survivorship Bias	Are failed cases, discontinued projects, or negative results missing from the analysis?
Publication Bias	Are positive results over-represented because negative results are less often published?

■ SECTION 07

RECURSIVE REFINEMENT FRAMEWORK

The system does not stop at first-pass output. It runs a structured self-correction loop that catches errors, fills gaps, and upgrades confidence scores before final delivery.

ITERATIVE DEEPENING PROTOCOL

PASS 1	BREADTH PASS	All 6 workstreams execute in parallel. Objective: establish baseline coverage across all domains. Depth: Layer 1 only.
↓ feeds into ↓		
PASS 2	DEPTH PASS	Command Core identifies the 3 highest-value findings from Pass 1. Agents re-investigate these at Layer 2 (mechanism level).
↓ feeds into ↓		
PASS 3	ADVERSARIAL PASS	Critical Challenge Agent attacks all Pass 2 conclusions. Any claim that fails receives a RED tag and is sent for re-investigation.
↓ feeds into ↓		
PASS 4	SYNTHESIS PASS	Strategic Synthesis Agent merges all passes. Resolves AMBER tags. Escalates RED tags to the executive summary as open questions.
↓ feeds into ↓		
PASS 5	COMPRESSION PASS	Precision Formatter compresses to all requested output tiers. Final quality gate check before delivery.

QUALITY GATES (ALL MUST PASS)

✓ Every factual claim has a source citation or [INFERRED] tag
✓ Confidence scores assigned to all conclusions (HIGH / MEDIUM / LOW)
✓ At least 1 contradicting source investigated per major claim
✓ At least 1 failure scenario documented per key finding
✓ No [CONFLICT-UNRESOLVED] tags suppressed — all visible in executive summary
✓ Geographic representation check passed (no single-region dominance without flagging)
✓ Critical Challenge Agent has reviewed 100% of synthesis outputs
✓ Ultra-compressed version contains the 5 single most important insights — nothing else

SECTION 08

IMPLEMENTATION GUIDE

STEP-BY-STEP DEPLOYMENT

STEP 1 Define your topic precisely

- Be specific: "AI chip supply chain vulnerabilities in Taiwan" beats "AI chips"
- Include a time boundary: avoids Kimi returning outdated information
- State your specific outcome: what are you building with this research?

STEP 2 Set stakes and audience

- HIGH stakes = maximum verification passes, 5-pass depth protocol
- MEDIUM stakes = 3-pass protocol, standard verification
- LOW stakes = 2-pass protocol, rapid summary output
- Audience determines jargon level and output abstraction

STEP 3 Copy the Master Prompt (Section 02)

- Replace all [BRACKETED] variables with your specific parameters
- Do not remove any Layer — each serves a structural purpose
- Add domain-specific workstreams if needed (replace A–F or add G, H)

STEP 4 Answer all clarifying questions

- Kimi K2.5 will ask 2–3 questions before initiating — answer completely
- Skipping questions = 100 agents operating on incorrect schema
- On Claude/GPT: explicitly state 'no clarifications needed, begin pass 1'

STEP 5 Request specific output format

- Specify by number: "Output Format: 03 + 09" for Deep Analysis + 1-Page
- Request multiple tiers simultaneously — Formatter handles all in one pass
- For presentations: request Format 06 (Strategic Implications) + Format 08 (Implementation)

TOPIC CUSTOMIZATION EXAMPLES

DOMAIN	EXAMPLE TOPIC	RECOMMENDED WORKSTREAMS
Finance	Emerging market sovereign debt risk 2025	Macro / Credit / Geopolitical / Regulatory / Historical Comparisons

Technology	Quantum computing commercialization timeline	Physics / Hardware / Software / Market / Competitive Landscape
Healthcare	GLP-1 agonist long-term cardiovascular effects	Clinical / Mechanism / Safety / Market / Regulatory / Competing Therapies
Geopolitics	Taiwan Strait escalation scenarios	Military / Economic / Diplomatic / Historical / Domestic Politics
Climate	Carbon capture technology readiness	Technical / Economic / Policy / Deployment / Environmental Impact

■ PRO TIP: For the highest-quality outputs, send this system prompt first, then in a separate message provide: 'TOPIC: [your topic]'. This forces the model to fully initialize the agent schema before receiving the research target.

SYSTEM IS NOW READY. PROVIDE YOUR RESEARCH TOPIC.

Replace [TOPIC] in the Master Prompt and deploy.